Recognising vessel movement from satellite image sequences

A beginner-friendly research brief | X48 | 10 November 2026

Use recorded ship paths to create training images, then test whether a model can recognise movement in real satellite footage without receiving AIS.

1. The problem and proposed idea

AIS (Automatic Identification System) broadcasts a vessel's identity, position and movement. A vessel may be visible in satellite imagery even when no reliable AIS match is available. This project asks what its **visible movement** can tell us.

Train a model using **synthetic image sequences**: computer-rendered views of ships following real AIS-recorded paths. At use time, a detector finds a vessel, a tracker follows it across frames, and a classifier assigns a movement label from the resulting evidence.

AIS tracks
Recorded paths
→
Synthetic frames
Rendered examples
→
Train a model
Learn movement

At use time: real satellite frames > detect > track > classify
AIS is absent from the prediction inputs.

2. What could it recognise?

Candidate output	Plain-language meaning and evidence needed
Transit	Sustained travel in a fairly consistent direction. Needs several positions over time.
Slow / near-stationary	Little displacement over a defined window. Does not by itself distinguish anchoring, drifting or mooring.
Turning	Direction changes across the observed sequence. Needs enough motion to exceed tracking noise.
Approaching another vessel	The distance between two tracked vessels decreases. Both must be visible and tracked together.
Loitering / zone entry *Optional extensions*	Loitering needs a sufficiently long observation; zone entry needs a known boundary. Include only if real test data support them.

Recommended starting scope
Choose only **2-3 classes after auditing real footage**, with an **unknown / insufficient evidence** option. A dark vessel means no reliable AIS match; movement labels can support investigation, but do not establish illegality or intent.

Optical imagery is camera-like imagery; SAR uses radar. This proposal starts with optical sequences. A sequence may be spaced frames rather than a smooth movie, provided its timing captures the behaviour.

What previous research already does

The linked studies below establish the main building blocks. A ship-type label such as yacht or tanker differs from a movement label such as turning. Results across different datasets are not directly comparable.

3. Vessel detection, tracking and type recognition

Study / link	Sensor input	Established capability
Mayra et al. (2025)	Sat-Optical Sentinel-2	Small/recreational vessel detection; **8,768 annotations**; test F1 **0.863**. Boxes can include wakes; performance is not a universal small-boat size guarantee.
Ciocarlan & Stoian (2021)	Sat-Optical Sentinel-2	Ship detection using self-supervised pretraining and limited labelled examples.
MP2Net (2024)	Sat-Optical video	Joint detection and multi-object tracking, including ships in SAT-MTB. Tracks identity; does not assign maritime behaviour classes.
SVSDet (2024)	Sat-Optical video	Fine-grained vessel-type recognition. Establishes ship categorisation from video, rather than the proposed motion labels.
xView3-SAR (2022)	Sat-SAR Sentinel-1	Vessel detection and characterisation in nearly **1,000 SAR scenes**. A snapshot benchmark, not a temporal behaviour dataset.

4. Movement states and activity-related patterns

Study / link	Sensor input	Outputs and important distinction
Chen et al. (2020)	AIS tracks	CNN classification: **static, normal navigation, manoeuvring**. AIS paths are already suitable for learning movement categories.
Zhang et al. (2022)	AIS tracks	Loitering detection and **4 trajectory-shape classes**; CNN shape accuracy **86.2%** on its own test.
CNN-Lite (2025)	AIS tracks	**Mooring, anchoring, sailing** from trajectory images; reported accuracy **91.54%**.
Pitsikalis et al. (2019)	AIS + spatial context / metadata	**Rule-based** loitering, rendezvous, drifting, anchored/moored and other events. Rules supply operational labels, not complete independent activity truth.
Cope et al. (2022)	Coastal radar tracks	**Focal** and **linear** movement patterns using speed/heading rules, associated with potential fishing.
Grasso (2019)	Sat-Optical Sentinel-2; AIS labels	Snapshot speed classification at a **1-knot threshold** and course quadrants. Speed experiment used ships **longer than 200 m**.
Ballinger (2024 preprint)	Sat-Optical PlanetScope; AIS check	Detects **2 ship-to-ship transfer configurations** in snapshots. Spatial event evidence differs from observing a complete encounter over time.

How training data would be made

5. From AIS positions to synthetic frames

1. Select real paths. Clean timestamped AIS positions, remove impossible jumps, and keep continuous segments. Separate vessels/voyages before generating samples. For two-vessel approaches, use genuinely simultaneous tracks from the same area.

2. Define labels. Apply explicit speed, direction-change, displacement or pair-distance rules to each time window. These are **weak labels**: a reproducible description of motion, with no independent confirmation of intent. Missing AIS intervals cannot be made factual by interpolation.

3. Render the scene. Place ship sprites or 3D models on sea backgrounds using the recorded positions and orientations. Blender, Unity or simple compositing can generate frames; **generative AI is optional**. Sample appearance consistently through a clip.

4. Match the intended sensor. Vary pixel scale, view angle, lighting, sea texture, blur, camera movement, haze and missed detections. For scale: a 30 m vessel is about 30 pixels long at 1 m/pixel, but only 3 pixels at 10 m/pixel. Preserve timestamps and realistic displacement.

5. Export and train. Save frames, boxes, vessel IDs and labels automatically. Start with a pretrained detector/tracker and a small sequence classifier. Cache tracks/features to reduce GPU use. A trajectory image used by an AIS CNN is a drawing of a path; it is not a realistic satellite scene.

6. Relevant synthetic-imagery precedents

Study / link	What it generated	What remains different here
UnityShip (2021)	Over **100,000** aerial ship images, from **79** ship models.	Still-image ship recognition; does not establish AIS-driven temporal behaviour transfer.
MarSyn (2022)	**25 videos x 1,000 frames** using Blender; automatic masks.	UAV-style viewpoint and segmentation task; useful rendering precedent.
RedNavi (2022)	3D maritime scenes driven by received AIS.	Navigation visualisation, rather than satellite-image behaviour training.

7. Candidate data and their roles

Dataset / link	Role and remaining check
Composite Maritime Events	AIS-derived rule events: training definitions/examples. Not paired with the satellite clips; underlying track access and alignment must be checked.
US Marine Cadastre AIS	Raw AIS paths for simulation. Requires cleaning and event labelling.
SAT-MTB	Real satellite video: **249 videos, >50,000 frames** across several object categories. Audit ship-only counts, durations, access and labels.
OOTB	Real tracking benchmark: **110 sequences, 29,890 frames** overall, including ships. Behaviour labels and usable event coverage still need checking.

Dataset totals above are not counts of independent ship events. No checked source yet secures a ready-made combination of real satellite ship video, matched AIS and verified labels for all proposed behaviours.

What would make this valid research?

8. The contribution to test

Research question: Does training with AIS-driven synthetic optical sequences improve movement recognition on held-out **real satellite footage**, compared with simple rules and models trained on noisy AIS coordinates?

The potential novelty is a measured **synthetic-to-real transfer** result: which rendering choices help, which behaviours remain visible at different resolutions and sampling intervals, and when the model should decline to answer. The checked sources do not establish this exact full pipeline; that is a **candidate gap, not a first-ever claim**.

Domain gap means rendered images and real images look different. **Domain randomisation** varies the renderer to reduce reliance on one appearance; **domain adaptation** adjusts learned features toward real data. Either is useful only if it improves an untouched real test.

9. Minimum evaluation plan

Requirement	Practical test
Real reference labels	Use each real clip's annotated positions to define observable motion labels. Check camera motion and tracking errors. Pixel distances are not metres without calibration; long-duration or geographic claims need suitable metadata.
Fair baselines	Compare: track rules; a compact classifier trained on corrupted AIS; and synthetic-image training. Also compare reference tracks with model-extracted tracks to expose detection/tracking failures.
Honest splits	Keep entire videos, overlapping source footage and vessel journeys in one partition. Separate training/calibration from final testing; use region holdouts if geographic transfer is claimed.
Useful measurements	Per-class precision/recall and macro-F1 (equal-weighted class balance); false event alerts; track failures; runtime and memory. Report intervals grouped by video/vessel, not by frame.
Controlled comparisons	Remove rendering or its degradations; vary frame interval, resolution and clip duration. This tests whether images add value beyond noisy coordinates.

A test against rule-derived labels measures agreement with that motion definition. Using the same rules on perfect reference tracks is an oracle check, not evidence of semantic understanding. Synthetic-only accuracy cannot establish real-image performance.

10. Feasibility, team roles and decision gate

Low-GPU design: bounded image crops, conventional rendering, cached detector outputs and a compact temporal model. Avoid a new large video generator or SAR simulator. Actual resource needs must be benchmarked in the pilot.

Three connected research roles: A studies sampling and behaviour observability; B compares visual/track models and tracking-error effects; C studies rendering choices, real-data transfer and confidence. Each needs its own baseline and experiment, beyond building software.

Conditional go: run a 2-3 week data pilot first
Download and inspect real ship clips; count independent examples for 2-3 visibly supported classes; establish reference labels and a simple baseline. Proceed only if a defensible real test is possible. If not, narrow the claim to simulation feasibility or redesign the task.

Limits to keep explicit: ordinary Sentinel-2 revisit-spaced snapshots do not reconstruct short manoeuvres. Synthetic frames cannot recover missing real observations. Hiding AIS during testing proves AIS-free inference on that sample; it does not prove accuracy for every small or AIS-silent vessel. The output supports screening, not a finding of illegal activity.